

A Novel Solution to Convert Between the USB 3.0 Protocol and Automotive
Ethernet at the Physical Layer

Capstone I Final Report



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1 Introduction

1.1 Project Introduction & Need Statement

As self-driving cars become more advanced, and as artificial intelligence brings new horizons to the automotive industry, the need for new technology in the automobile industry rises. In particular, cars are becoming more complicated systems, akin to computer networks of devices all linked up together and sharing information. Especially in self-driving cars, this information needs to be shared quickly and efficiently, with easy to use standards and protocols. The standard for the physical (electrical) layer of networking is the ethernet protocol, the system in which each cable contains four differential pairs to ensure redundancy in the bidirectional communication. For car manufacturers, however, the electrical design of ethernet cables poses a problem, as vehicles must be optimized for weight and power consumption.

Thus a new standard was needed, adapted from the original model of ethernet, to handle the same data transfer capabilities and speeds as ethernet, but with less wire management. Automotive Ethernet[1] is such a solution, capable of high speed data transfer with just a single pair of wires. Compatibility with the original ethernet is almost always standard now, with almost every device capable of communication over the ethernet physical layer. However, due to the nature of Automotive Ethernet, compatibility would require extra electronics manufacturing to transform ethernet signals into those readable by an automotive ethernet device.

In particular, the network-like characteristics of internal car communications made automotive ethernet the standard for self-driving automobiles, but difficulty arises when users must interface with the network to perform diagnostics or verify communication. Currently, there is no inexpensive way to transform an automotive ethernet signal to one readable by a computer, such as USB[2, 3], meaning the data transfer requires a sequence of adapters, which could hinder speed and productivity. Our project aims to address this need for such a device.

1.1.1 Framed Insights & Opportunities

The client requires an all-in-one device for communicating with a car. The device should have a female USB-C connector for connecting to a computer, and a female “H-MTD” connector for connecting to the car. The device should communicate with the car using automotive ethernet, specifically “1000BASE-T1”[4]. The device should have status lights that indicate connection status and speed. Ideally, it should be capable of gigabit speeds. The device should have jumpers or switches to select between a “master” or “slave” mode. It will probably be used to do things like read vehicle trouble codes and review information that was collected by the vehicle during a drive, like the paths that it took, time of events, like when it detected certain objects, and values of various sensors. However, this product could be useful to many more people than just Zoox employees.

Some stakeholders, such as certain modern electric vehicle companies, don’t use OBD-2 ports, and have their own proprietary connectors and communicate over ethernet. With some minor modifications, this device could probably be used to service other vehicles, like those made by Tesla or Rivian. If this device does end up getting made to work with any suitable make and model of car, it might use a screw terminal instead of an H-MTD connector so that the end user can use any connector that they want. After all, plain wires are the universal adapter. A device meant for an end user should probably be more than just one or two PCBs. If this device were made available to the broader community, it should probably have a sturdy case, and a marked panel that says what each port is for, status lights, and switches to configure anything that can’t be set up automatically. It would get annoying very quickly if a user had to take apart the device to make changes.

1.2 Client & Stakeholders

This project was proposed by Austin Hendrix, an employee at a company named Zoox. Zoox is a start-up autonomous vehicle company founded in 2014. Their mission is to develop versatile autonomous vehicles accessible to the wider community. They aim to “make personal transportation safer, cleaner, and more enjoyable - for everyone”. In service of this goal, they have developed a ride-hailing vehicle (in essence, a taxi) designed for urban landscapes and environments. By tailoring the vehicle’s artificial intelligence to the specific area of operation, the company is able to produce very powerful and advanced results.

Zoox plans to develop a vehicle from the ground up, including software, hardware, and AI, and in the long run hopes to reduce the need for personal car ownership and minimize environmental impact. Due to the nature of this design philosophy, the company includes many layers of development including manufacturing, engineering, and design. Every layer of the development has a stake in the project outlined in this document, as diagnosing problems and ensuring proper behavior must be done at every stage of automobile creation. Manufacturers need to verify communication, engineers need to debug new functionality, and every developer needs to be able to interface with the car. Thus it is incredibly important to transform signals from USB to automotive ethernet. Currently, Zoox uses a setup of cascading adapters (adapters wired into one another in sequence) to convert between the two electrical standards, but we aim to accomplish this in one self-contained device.

In addition to Zoox, this product may be of interest to the wider automobile community, as it has the potential to aid in more advanced car development and to streamline the verification process for many different companies. If this device, in later revisions, is made portable to different connectors and USB standards, it is possible to be of use in other vehicles. One interesting stakeholder group may be farmers, or farm workers, as vehicles like tractors need to be fixed and verified, and as industrial farming becomes more modernized, those vehicles become more advanced, and this product could be an inexpensive way to diagnose problems in more advanced industrial automobiles including tractors and similar types.

This product provides the user with a more portable and compact way of interfacing with cars that use automotive ethernet for inter-device communication, and therefore could be of interest to the automobile community as a whole.

1.3 Project Objectives

The main objective of this project is to create a functional adapter that transfers data between automotive ethernet and USB. This will serve as a proof-of-concept demonstrating that the transfer can be achieved in a single compact device. To accomplish this task, a series of project objectives was created to provide a set of goals and milestones to complete in order to achieve the main objective. This section aims to introduce these objectives and describe how they contribute to the overarching project.

First, the team must document the USB and automotive ethernet standards, including those for regular ethernet, and interface protocols such as RGMII and SGMII. This is a necessary first step, as the team needs to be knowledgeable enough to consider all angles and to diagnose potential problems in the future. In addition, the standards will form a library for future reference while working on the technical side of the project.

Second, a decision must be reached on a main design for the project. There are many distinct ways of accomplishing the functionality required for this device (see appendix A) but the team must decide on a high-level method in order to proceed with the development. This includes the kind of devices or chips needed, which connectors will be used, and the overall transformation scheme of the

device.

Third, the team must debate the specific chips and/or hardware for the device. Once an overall design is chosen, there will be many different types of integrated circuits or manufactured chips to choose from, all of which will provide the required functionality. The team must reach a decision about which specific electrical components to use, so that the device can be made properly and in a timely manner.

Fourth, the PCB must be manufactured. It has been agreed that this device, no matter which design is chosen, will require a custom PCB to be built. The choice of which company, how many layers, and which board layout to use will be an essential step in finishing this project.

Fifth, the team must verify that the created device works in the intended fashion. The tests must be designed, built and defended for accuracy, and the device must be verified to work. This includes tests for data transfer, to ensure the data is being converted correctly, and tests to push the device and test the physical limits of the adapter. Performance testing is always necessary, especially to verify speed and data size compatibility, as well as providing the client with recommendations and maximum safety values.

Finally, the device must be packaged appropriately and given all of the required user interface components. This includes LEDs, switches and other related packaging that the customer will interact with. Once the adapter has been verified to be working, these extra components can then be added onto the device to finish the project. Our main goal with this project is to provide a working product at the speeds required by our client. In addition to this, we want to achieve a great deal of learning and gain a good amount of experience by working on this type of device.

1.4 Project Deliverables

The main deliverable is an adapter with the purpose of conversion between Automotive Ethernet[1] and USB-C[2, 3], enabling easy data transfer between modern automotive systems and other devices. The adapter will play a vital role in the engineering process and will make troubleshooting and testing less cumbersome for the production and engineering teams at Zoxx. While form is not our top priority for this phase, the proof of concept will help cement the fact that our method works reliably in its intended use environments. Our hope is that the completed adapter prototype will be the first step in producing a tool to be used in many different automotive applications.

The team aims to provide this device in a functional form, meaning the prototype works according to the specifications, with the required speed and data capabilities as stated by the client. This project is a standalone device, and therefore will require documentation concerning operation and recommended use conditions, which will be provided by the team.

1.5 Team Introduction

Our team, appropriately titled Team-Mobile, is composed of four individuals, each with a unique skill set and specific roles for this project. First, Alex Short is acting as the project manager for this project, in charge of running meetings and keeping the team organized. This project has two major components, the hardware design and fabrication of the device itself, and the software architecture used to test and verify the device. With this in mind, roles were delegated to each member based on preference and skill set. The system architect, in charge of hardware design, is Vladimir Gupenko. He has worked hard on the current alpha product to test the viability of certain chips. Next, Benjamin Lappen and Seth Abramowitz are the software architects and designers. These roles are mostly concerned with higher level electrical design and developing any testing software needed

for verification. In addition, they will work on any microcontroller programming necessary for the project. Alex Short will work on hardware and electrical design, as well as project planning.

Our mission as a team is to work alongside one another to deliver a satisfactory product to the client. We need to be organized and on top of development, documentation, and production. The team should be split into well suited roles and jobs that interconnect and work together in effective ways to accomplish the project goals. Perhaps most importantly, we need to be willing and open to learn new skills and technical knowledge that will aid in the design and manufacturing process. In addition, we need to be open to considering unique ideas and examining extra points of view in order to develop the best product possible.

1.6 Project Duration and Planning

This project is planned to begin in the Fall of 2024 and conclude in the Spring of 2025. It is important to create a plan that encompasses the entire duration of the project including important dates and milestones. The milestones we believed to represent our project are as follows:

1. 10/25/24 - Research the Problem and Stakeholders

The first task is to research the problem, which includes gathering documentation and learning the necessary electrical design skills for this project. In addition, it is important to research and understand which groups may have a stake in this project so that the final design can be as close to the customers' usage needs as possible.

2. 10/25/24 - Develop initial requirements for a solution

After the team has learned the materials necessary to complete this design, the next step is to develop specific requirements for a solution. Creating a clear picture of a finished design will allow the team to more efficiently brainstorm ideas for the product.

3. 11/1/24 - Propose Solutions that Meet Requirements

This step encompasses the brainstorming process for the project. This milestone represents the transition from the brainstorming process to the concept selection process. The proposed solutions will need to satisfy the initial requirements laid out at milestone two.

4. 11/8/24 - Concept Selection

Clearly, after the brainstorming process is completed, the best idea must be chosen to be the main design for the project. This will require discussion and reflection, and potentially debate. The team must decide on a design idea that will serve as the basis for the device developed during the course of this project.

5. 11/16/24 - System Diagrams

Both high level and low level system diagrams are required to represent the device operation and logic. The electrical diagrams need to account for power consumption and different voltage requirements, while the higher level diagrams need to depict the flow of data in a logical and easy-to-understand chart.

6. 01/2025 - Complete Product Testing

It will be necessary to have completed an early testing and verification process on a manufactured design by January of 2025 in order to have time to perform revisions and fixes on the design if necessary. This will require PCB design to be completed by early January.

7. 02/2025 - Next Iteration

The next month, February 2025, will need a second revision to be completed as more fixes may be needed, or a fix that was applied in January may need to be verified.

8. 03/2025 - Completed Device

The final milestone is the completed device. This project is scheduled to end in the March of 2025 and therefore a completed product is needed by then.

This project requires a good deal of planning, as PCB manufacturing and design requires time. In order for discussion to be productive, team members will need an understanding of the relevant protocols early on in the brainstorming phase. Those protocols include:

- Automotive Ethernet (1000Base-T1)
- Ethernet (1000Base-T)
- USB 3.0
- RGMII
- SGMII

In addition, designers will have to verify that the design meets impedance specifications for each communication standard, and our stakeholder desires some ability to see the current transfer link speed and activity physically on the device, which will need to be factored into any design. Finally, the connectors for both ends will need to be found, and automotive ethernet parts may be difficult to acquire, as they are not used widely in most industries.

2 Related Works

As we developed the engineering behind this project, it was helpful to draw inspiration from many different related sources. The past work done by engineers and scientists has contributed greatly to this project. Due to the nature of this project we largely examined other adapters and how they were designed as a benchmark for our more complicated device.

2.1 USB to Ethernet Adapters

USB to Ethernet adapters[5] are a common and widely available solution for providing wired network connectivity to devices without an Ethernet port. These adapters typically use USB 3.0 or 2.0 for interfacing with a host device, offering speeds up to 5 Gbps (USB 3.0) and supporting gigabit Ethernet protocols.

Their design principles, compact size, and plug-and-play functionality provided inspiration for the USB 3 to Automotive Ethernet adapter. The straightforward user experience of USB-to-Ethernet adapters influenced our goal to make our Automotive Ethernet adapter equally simple to use, ensuring compatibility across operating systems such as Windows and Linux. However, the transition to Automotive Ethernet introduced challenges such as dealing with different physical layer protocols, higher data rates, and stringent automotive standards.

The design of USB adapters provided necessary insight into the electrical workings of USB data transmission at the physical layer. This was needed to understand how to properly take the USB signal and transport it through the PCB layout.

2.2 Automotive Ethernet Adapters

Automotive Ethernet adapters[6] are specialized devices primarily used for development, diagnostics, and testing within automotive ecosystems. These solutions often rely on proprietary hardware or software stacks, tailored to meet the specific requirements of automotive manufacturers. They typically interface with standard Ethernet for backward compatibility but are not designed with compact, consumer-friendly form factors or cross-platform support in mind. Common limitations include high costs due to specialized hardware, restricted accessibility for general-purpose developers, and poor integration with consumer-grade tools like laptops or PCs.

Automotive Ethernet-to-Ethernet adapters, the most common type of Automotive Ethernet adapter, play a critical role in environments requiring data translation between standard Ethernet and Automotive Ethernet. These devices enable seamless communication between traditional network systems and modern, high-speed, full-duplex Automotive Ethernet, essential for vehicle diagnostics and integration with external systems.

Our project leverages the principles of these adapters for designing the Automotive Ethernet-to-RGMII (Reduced Gigabit Media Independent Interface) component of the USB 3 to Automotive Ethernet adapter. By combining the strengths of both types of existing solutions, our design aims to bridge the gap between Automotive Ethernet and conventional networking hardware, ensuring low latency, high reliability, and an accessible, affordable solution for broader use.

There is limited information on the electrical characteristics[7] of Automotive Ethernet, and the design of different physical layer interfaces and adapters for this protocol aided in the understanding of many different properties of such devices. As our own device falls under this category, having different examples was very helpful.

2.3 Our Project

The insights gained from analyzing these devices shaped the key objectives of our USB 3 to Automotive Ethernet adapter project:

- **Ease of Use:** Inspired by USB-to-Ethernet adapters, our device must provide a plug-and-play solution without requiring extensive driver configuration.
- **Affordability and Accessibility:** By addressing the cost and usability barriers of current Automotive Ethernet adapters, we aim to make the technology accessible to developers, hobbyists, and automotive engineers.
- **High Performance:** Leveraging the high-speed capabilities of USB 3.0 and the advanced features of Automotive Ethernet, our device must offer reliable data transfer suitable for automotive applications, including diagnostics, testing, and data logging.
- **Cross-Platform Compatibility:** Building drivers for Windows, Linux, and macOS ensures the device works seamlessly across various environments, a limitation in many existing solutions.

The resulting adapter will bridge the gap between consumer and automotive technologies, addressing the growing need for accessible Automotive Ethernet tools in modern development workflows.

3 Project Overview

This project is a multi-stage engineering development project aimed at designing a specific device to accomplish a certain functionality. The Automotive industry requires the use of network protocols to communicate between devices, and we are designing an adapter to interface with such networks. The adapter will transfer data between a USB 3.0[2] connector and an Automotive Ethernet[1] connector.

As we developed the project, we created different sets of requirements and constraints to outline the system design. The requirements were derived directly from interactions with the client and the properties of automotive ethernet, and then translated into more engineering-focused requirements to aid in the low-level hardware design.

3.1 Marketing & Customer Requirements

The requirements put forward by our client were to describe the creation of an all-in-one device for communicating with a car network using Automotive Ethernet as the protocol. These requirements, called marketing (or customer) requirements, describe the functionality of the device from the perspective of the user. The device is meant to be a proof-of-concept that such an adapter can be accomplished in a single device, meaning these requirements are much more focused on functional behavior than the look and feel of the user experience.

The main customer requirement is of course that the device transfers data between a USB connector and an Automotive Ethernet connector without losing packets or corrupting the data in any way. Ideally, this device should be capable of link speeds of 1 Gbps (Gigabit) and operate using the 1000BASE-T1 specification[4].

In addition, the device should be compatible with many different users, such as both Linux and Windows operating systems, so that the adapter is flexible enough to be used by many different technicians and engineers. In addition, this flexibility means that the device should be easy to transfer and to operate, without requiring very much setup. On regular Ethernet ports, there are LED lights that blink on and off corresponding to the link speed and the stability of the connection. Our device should have similar lights to mirror a regular ethernet adapter and get as much information to users and testers as possible.

The complicated nature of the automotive ethernet physical layers requires that certain devices are in a 'controller' or 'peripheral' state during different modes of operation[8]. For the most flexible device possible, there should be a switch controlling the mode of operation, allowing users to troubleshoot and tailor the adapter to their needs. Another necessary switch is one that controls the link speed, ideally between 100 Megabit and 1 Gigabit speeds to portability to all kinds of vehicles.

Zoox has expressed interest in mass producing these devices internally for a variety of users. This implies the cost should be low enough to support production on a larger scale within a company.

Finally, there were two customer constraints on the device hardware itself. As the Zoox cars are designed in a certain way, our client specified that both ends of the device should be female connectors for the respective protocols. One end of the device should be a female Automotive Ethernet connector, and the other end should be a female USB-C[3] connector.

In summary, there are a set of marketing requirements that describe the functionality of the device from the user perspective, and a couple of constraints on the device hardware itself that will restrict the potential solutions the team is able to create.

- The device should communicate via a USB link to a Automotive Ethernet network, ideally under the 1000BASE-T1 specification, with Gigabit speeds
- The device should be portable and available to a variety of users

- The device should display status information us LED lights, such as link speed and stability
- The device should account for switching between controller and peripheral modes of communication
- The device should account for switching between distinct link speeds
- The device should be producable on large scales without an incredible cost
- The connector for the computer-side of the adapter should be a female USB-C connector
- The connector for the vehicle-side of the adapter should be a female Automotive Ethernet connector

3.2 Engineering Requirements

These marketing requirements discussed in the previous section lead to the development of engineering requirements, measurable requirements for the device functionality that are needed to achieve the customer requirements.

ID	Parameter	Requirements	Tolerance	Risk	Compliance
1	Transmission Speed	1 Gbps	Max.	High	Test
2	Connector Layout	female-to-female	N/A	Low	Inspect.
3	Self-Contained	0 external power need	Max.	Low	Test
4	Drivers	Windows & Linux	N/A	Medium	Test
5	Display Lights	Status & Speed	Min.	Medium	Test
6	Base Functionality	0 Losses	Min.	Low	Test
7	Mode Switch	Successful Mode Switch	Min.	Low	Test
8	Cost	Under \$150	Max.	Medium	Inspect.

Table 1: Engineering Requirements

The compliance method dictates how we will verify the engineering requirement is met. Inspection means we will examine the device to make sure it fulfills the requirement, and testing means we will need to systematically verify that the requirement is met.

A few of the engineering requirements are directly taken from the marketing requirements. The customer requires a transmission speed of 1 Gbps which directly translates to a measurable value requirement. The risk for this one is high because it may be difficult to achieve such high speeds without many revisions of circuit board, which we may not be able to do in the project duration.

The connector layout was specified to be female-to-female, which we will need to include when developing designs, as this will affect the circuitry layout. This is low risk as it is just a change to the board layout and not dependent on the strength of the device. This requirement can just be verified by inspection.

One very important requirement is that the device operates on no external power draw. This means the device is self-contained and does not require any external cables which makes the adapter very portable and easy to use. This requirement is derived from the customer constraint that the device be available to a variety of users, as powering the device only from the USB connection means a user will not have to set up any external power. This will need to be verified by testing the device using only a USB connection.

Another way to make the device portable to a variety of users is to provide support for multiple operating systems, such as both Windows and Linux. This means that drivers must be present on both the operating systems for whichever USB PHY we choose to interface with the device. Testing this on both operating systems will verify that the device works.

As specified in the marketing requirements, the device must be mass producible internally for Zoox, meaning tens, potentially hundreds of these devices may need to be made throughout the departments. This means the cost of all the components and the PCB manufacturing needs to be low enough to support this without costing too much money. We felt a reasonable value was under \$150 for the entire process, without buying chips or components in bulk, as prices are often lowered when purchasing in bulk.

3.3 Design Criteria

In order to evaluate any potential designs with regard to compliance under the marketing and engineering requirements, the team developed a set of design criteria to use in ranking design concepts.

These criteria are as follows:

- Cost
- Time
- Research
- Reproducibility

Each of these corresponds to a core aspect of this project. How expensive are the components for the device? How much time will it take to complete? Can we complete the device in our project time frame? How much learning will this require? Will we be able to learn all the required skills for this? Will our client be able to easily reproduce this device on the scales they would like to?

These criteria will be used to evaluate the potential design solutions in order to narrow down ideas and select the best concept for the final system design.

4 Project Development & Planning

4.1 Concept Generation & Selection

By the time we reached the concept generation phase of our project, we had already developed a preliminary solution for the problem presented by Zoox. In this phase, however, we focused on exploring a broader range of ideas that could potentially address the issue in innovative ways. To facilitate this process, we leveraged ChatGPT’s brainstorming capabilities to generate fresh concepts. Each team member then selected one of these new ideas to research further, evaluating its feasibility for solving the problem. Our research included analyzing the pros and cons of each approach, estimating costs, and documenting additional considerations that might inform our final direction.

Ideation and concept generation were facilitated by multiple discussion activities in which the team members proposed and debated different solutions for the final device. In order to generate the ideas, we considered different technical manuals to understand how connections between the two protocols could be formed, and also used artificial intelligence tools such as ChatGPT to generate a few concepts. These concepts were then discussed and worked out in more detail before comparing the ideas all together. (See appendix A)

Out of these discussions, four ideas emerged as the top concepts for the final project. The first was a set of cascaded chips mounted on a PCB, meaning a highly electrical and circuitry focused design. Second, it was proposed to use an FPGA, by programming the digital logic corresponding to the transformations between timings, voltages, and data. The third idea was an embedded solution, potentially programming a microcontroller chip to carry out the data transformation. Finally, it may have been possible to use an SoC based design, in which the system is programmed to transform the incoming data to whichever protocol, using the peripherals and operating system of the chip.

To debate these ideas, three activities took place, using three different discussion strategies to ensure the debate was productive. The first technique was called the “Nominal Group Technique” which involved each person developing an individual list of ideas for the project. Then each person would present their best in turn and receive feedback. This started the ideation process and was the first debate about which idea may be the best.

The second technique was called the “Yes and...” technique. This method is a style of improvised conversation in which participants aim to add onto existing thoughts. This is helpful in facilitating the discussion as it prevents ideas from being shut down immediately and instead helps them grow into better versions. This showed the team which ideas were feasible and how they could be improved.

Finally, the last technique was called ‘SCAMPER’, which is an acronym which stands for Substitute, Combine, Adapt, Modify, Put to another use, Eliminate (or Elaborate), and Reverse. Each of these steps in the process allowed the team to perfect the ideas so that they could be evaluated against each other. In particular, the Adapt step was useful as it provided insight into the ways the various ideas could be amended for improvement, and it demonstrated the potential of each idea.

4.1.1 Concept Selection

Each of the four main ideas (see appendix A) were evaluated against the design criteria (section 3.2) to find the best idea. These ideas were fit into a design matrix to numerically quantify the "goodness" of each idea with respect to the design criteria. Appropriate weights were chose for each criterion so that the resultant score would best fit our project.

First, the cost of the design was weighted at 40% of the score, as the larger scale production needed by the company is very important, and the ideas presented here largely differ in cost alone.

Next, the other three criteria were all weighted at 20% of the final scores. This is because they are all important, but none were significantly more important than the others for our design.

Criteria	Weight	#1 Cascaded ICs	#2 FPGA	#3 Embedded	#4 SoC-Based
Cost	0.4	8	6	4	N/A
Time	0.4	8	2	3	N/A
Learning	0.4	5	5	5	N/A
Reproduce	0.4	8	3	5	N/A
Reqs. Met	Y/N	Y	Y	Y	N
	Score:	7.5	4	4.25	N/A

Table 2: Design Criteria Evaluation Matrix

Each score was out of a maximum score of 10, and the scores in each category were totaled and then averaged for the final score. A higher score for the cost criteria indicates how inexpensive the device will be, meaning the Cascaded ICs idea will cost the **least**. Again, a higher score in the time category indicates the least time required, and this applies for the the rest of the criteria except for reproducibility. In addition to the criteria, the ideas were checked to make sure they met all of the constraints from the client and were disqualified if not. The SoC-Based Solution, despite simplifying integration, was disqualified as it does not meet all requirements given by our client, mainly due to its high cost and limited availability.

The Cascaded ICs approach, involving Automotive Ethernet to standard Ethernet conversion followed by an Ethernet-to-USB adapter, scored the highest at 7.5/10 due to its cost-effectiveness, feasibility within the timeline, and ease of reproducibility. However, it introduces some latency and complexity. The FPGA-Based Solution, using custom IP cores, and the Embedded Solution, involving devices like ESP32 or Raspberry Pi, scored 4/10 and 4.25/10, respectively, due to higher complexity, development challenges, and reduced feasibility in the given timeframe. While both meet requirements, they are less practical for this project. Overall, the Cascaded ICs approach is the strongest option, balancing cost, feasibility, and reproducibility effectively.

4.2 Project Management

The integration of an Automotive Ethernet-to-USB adapter into a single, streamlined device presents the primary challenge of this project. Translating between Automotive Ethernet and USB protocols is complex, as they differ in structure, timing, and electrical requirements. Unlike standard Ethernet, Automotive Ethernet uses a single wire pair, demanding a robust translation layer that maintains data integrity and meets strict timing standards crucial for automotive performance. Additionally, weight and space constraints in automotive applications require a compact, efficient design that consolidates these functions into one device, reducing overall bulk but also intensifying precision demands in both hardware and layout to prevent interference and ensure durability. Synchronizing firmware and hardware across these interfaces adds complexity, as each protocol requires dedicated logic for initialization and data handling to prevent latency issues. This project requires expertise in many different areas: automotive communications, USB protocols, and embedded systems, all of which must come together to deliver a cohesive adapter. While the individual components are independently simple, their integration into a single, cohesive adapter is the project’s long pole, demanding substantial time and testing.

As stated, the long pole task is to combine the multiple chips onto a single board, accounting for different electrical and timing requirements. This is the most risky task as a single problem

with the electrical wiring will cause the entire board to become non-functional, and possibly damage components as well. Our plan to address this risk is to be as thorough as possible when examining and testing the device. When we receive a newly manufactured PCB board, the first step is to go over all the electrical and power related components, testing each route for short circuits and making sure power is routed exactly where it should be. Then the PCB designs themselves can be built in such a way to separate the various components from each other spatially on the board, allowing us to diagnose and separate out problems. It is not efficient for a single short circuit to make the board useless, so as a contingency we have installed many jumper pins through wires and in the board designs so that we can connect and disconnect routes as needed, or provide external power if the internal power planes are not functional. This is so that the PCB itself is robust and not susceptible to a single error during testing.

4.2.1 Project Scheduling

This project requires a good deal of scheduling in order to accomplish tasks on time and complete the project according to the planned finishing time. One way to organize this scheduling process is in a GANTT chart, which displays tasks in calendar like fashion. See figure 1 for our GANTT chart.

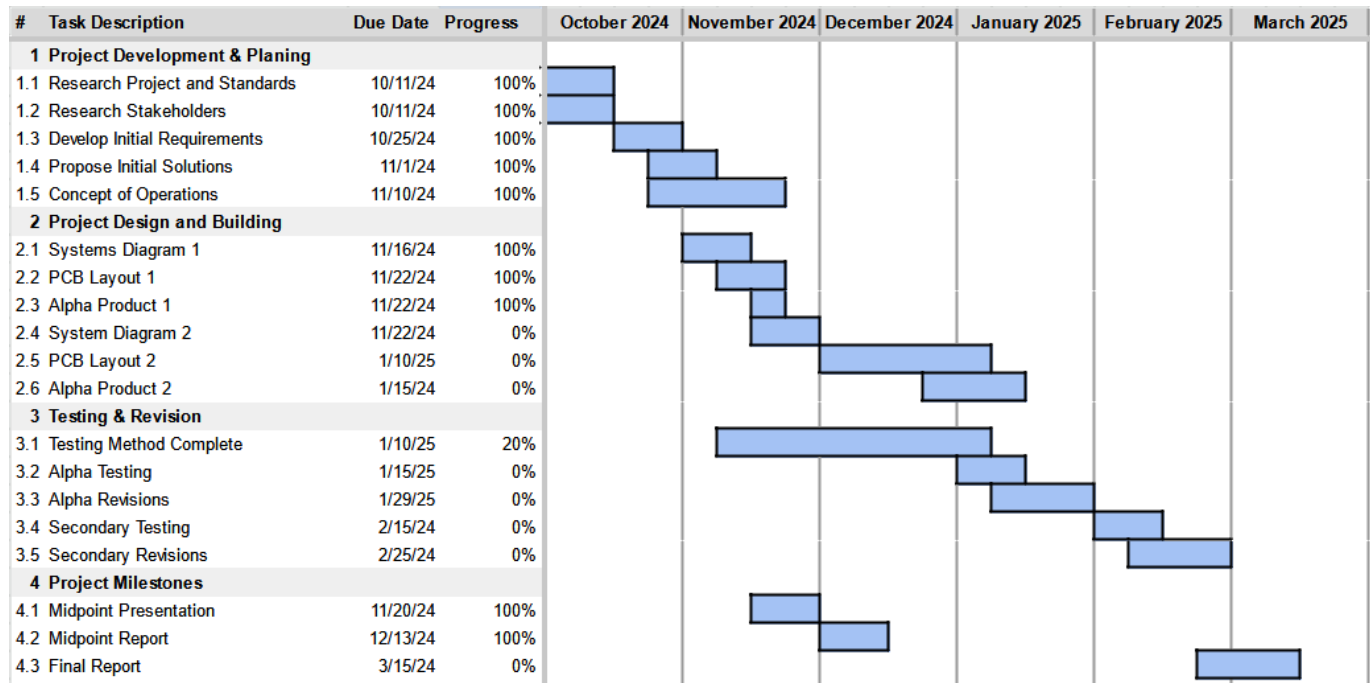


Figure 1: GANTT chart detailing the schedule and duration for the Zoox project

5 Alpha Products

We have been developing this device for a couple of months and have made progress on the designs for the PCB layout[7–9] and schematics. Each systems diagram we made uses the Cascading ICs approach to transfer the signal. In this design, two to three chips are placed in sequence, first transforming the USB signal to Ethernet, then Ethernet to an interface protocol such as RGMII or SGMII, and then that protocol to Automotive Ethernet. This section aims to break down our current designs and prototypes to demonstrate our current progress.

5.1 Design Version # 1

The first version of the adapter that exists in real life is a USB to standard ethernet adapter. The goal of this first design was to get parts onto a PCB and into a working state as quickly as reasonably possible. The main ideas for the PCB design were that if something is wrong, it’s best to find out sooner rather than later, and that the design should be forgiving of mistakes.

It takes a long time to fabricate the PCB and assemble the device, and because each adapter costs a non-negligible amount of money, it’s important to make the PCBs as configurable as possible, and for functioning portions to be capable of being tested independently of the possibly malfunctioning portions. That’s why these designs currently use plenty of jumpers and pin headers. If, for instance, there was a short circuit in the 5V portion of the board, the voltage regulator can be disconnected from the 3.3V portion, and power can be provided using a DC power supply instead. There are also jumpers for changing the pinout of the USB-C port, in case it turns out to be wrong. There are pin headers for programming the EEPROM, and there is a jumper for disconnecting power from the EEPROM. This is also useful for in-circuit programming, so that the programmer doesn’t try to power the entire board, and this makes the ethernet adapter effectively have two possible configurations.

For the alpha demo, we showed a computer using the USB to standard ethernet adapter to connect to the internet. The adapter is currently capable of transferring data at around 40 megabytes per second. This is better than 100-megabit “fast” ethernet, but it doesn’t reach gigabit speeds.

The experience of drawing the schematics in KiCAD was very useful. The LAN7800 chip has multiple connections to itself, and one component can be connected to many others in parallel. One of the lessons learned is how to draw complex schematics without them becoming an unreadable mess of connections.

This first version of the board helped show various shortcomings of the design, like the LAN7800’s sub-optimal default configuration (which defaults to its USB-2 mode), the long distances between capacitors and voltage sources, inefficient data trace routing, and inadequate power traces and grounding. Although it is true that plain wires can have a remarkably high bandwidth, even without any optimizations, the same is not necessarily true for PCB traces. Since the traces are often very close to each other, and they are located on more or less one plane, they require much more careful routing in order to get the performance that is required for this project.

The next version of the PCB will contain the USB to ethernet, and the automotive ethernet components. It will include the lessons learned from the first board, as well as from researching the topic of networking hardware.

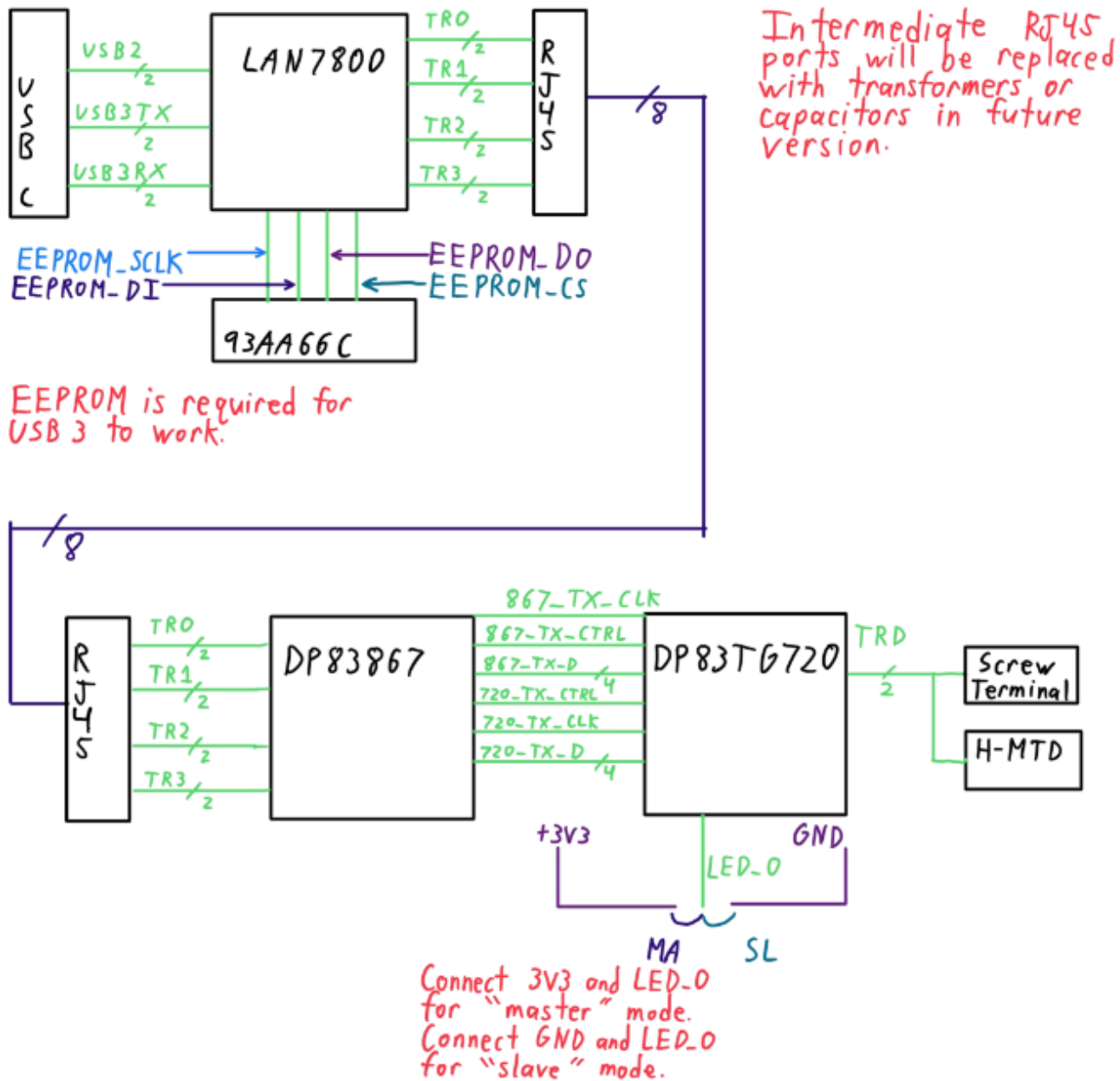


Figure 2: Block Design for Version #1 of the Device

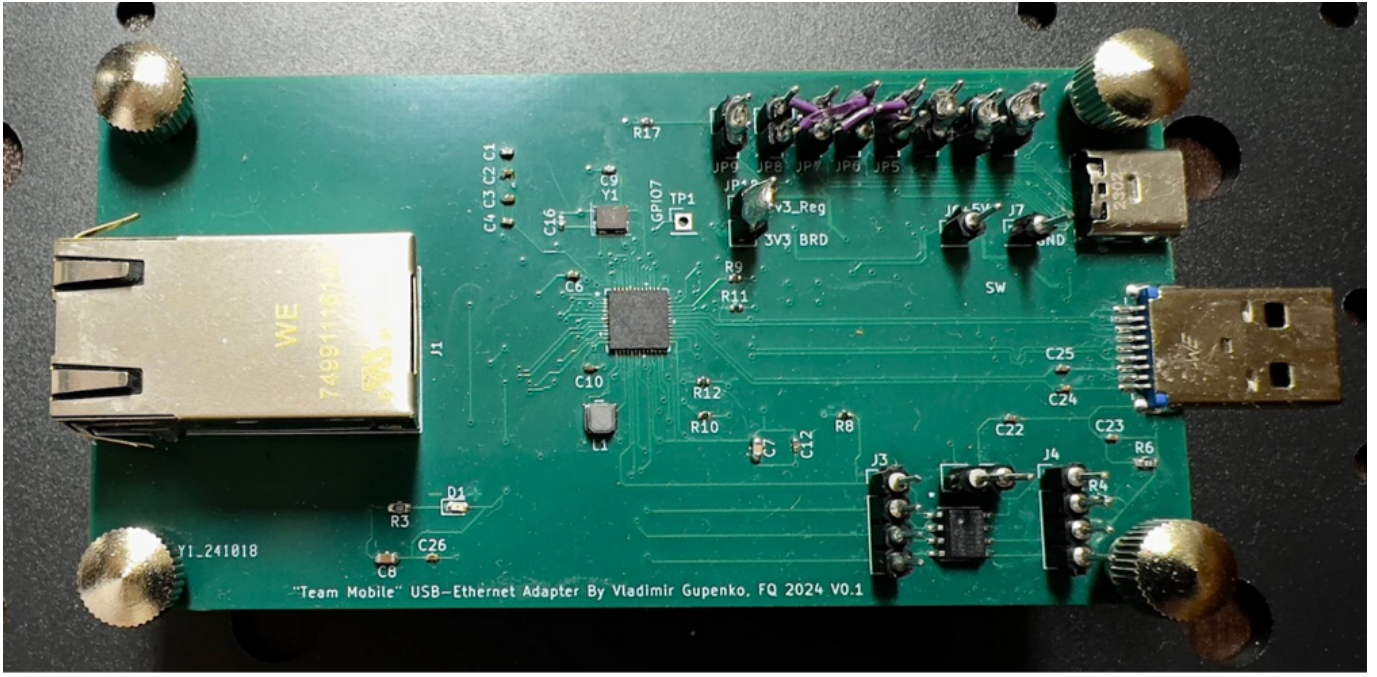


Figure 3: Manufactured PCB design of USB-3.0 to Regular Ethernet for Demonstration Purposes

The device pictured above is the first PCB we have manufactured in order to test the chips and routing layouts, as well as address potential problems as early as possible. Figure 2 is high level block diagram for the device including all the chips and data connections, without the power grid layout. The high level functionality is as follows. The USB signal is transported into the LAN7800[10] chip which converts the signal into Ethernet. In order for compatibility with USB 3.0, the LAN7800 requires settings to be loaded from an EEPROM[11], which is depicted in the block diagram. The RJ45 ports are placeholders for the power controls (transformers and capacitors) and transfer the Ethernet signal into the DP83867[12] chip which transforms the signal into one of the interface protocols, RGMII[13] or SGMII[14]. For this design we chose RGMII for its simplicity and portability.

5.2 Design Version # 2

Additionally, the team is pursuing an alternate design in parallel. The aim is to produce this adapter using the Microchip LAN7801[15], which translates USB directly into RGMII. By utilizing this chip, the overall design can be simplified to have only one Ethernet PHY rather than two. However, this system is in an earlier design phase, making it advantageous to still pursue our primary design.

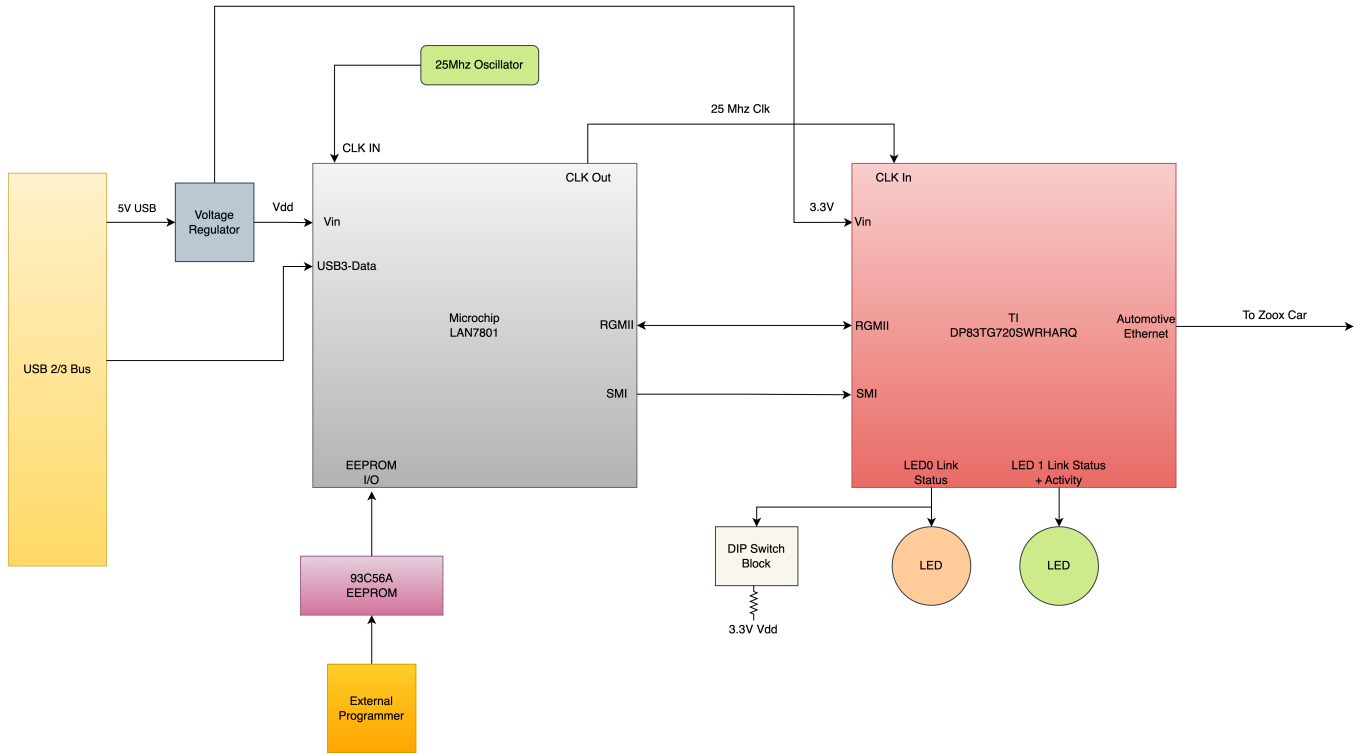


Figure 4: Block design for the second version of the device, using less chips

This design has the benefit of using less chips, two instead of three, and with a very similar cost. It will reduce the complexity of the PCB and schematics. Since we are still laying out the PCB and wiring the schematic, it is still advantageous to pursue the first version, which is much closer to yielding a complete product.

5.3 Current Product

Currently, we have the PCB shown in figure 3. A demonstration of the device in operation can be found at the URL:

https://drive.google.com/file/d/1STA81L8GJt83PUzIqTaIOvC4jS9uMFth/view?usp=drive_link

This demonstration shows the device being used to connect a regular laptop to the internet, showing the adapter is functional.

6 Teamwork

ITP Metrics is an online assessment tool designed to promote effective teamwork. By taking a survey, ITP Metrics is able to produce insights about the team dynamics, individual contributions, and overall team health. Furthermore, using this tool empowers teams to identify strengths, weaknesses, and areas for improvement. Reflecting on team results through ITP Metrics fosters open communication, encourages constructive feedback, and ultimately leads to enhanced collaboration and productivity. By understanding the team's performance and addressing any issues proactively, teams can optimize their processes, achieve better outcomes, and build stronger, more resilient relationships.

The team's peer review reflections (see appendix B) indicate a positive trajectory in team dynamics and individual performance. Initially, some members, like Alex and Seth, exhibited self-doubt and a need for improved focus. However, through consistent feedback and self-reflection, these areas have shown significant improvement. Vlad's scores have consistently been strong, demonstrating a solid contribution to the team. Ben's performance has also steadily progressed, with particular growth in focus and commitment. Overall, the team's collective effort and willingness to embrace feedback have resulted in a more cohesive and effective working environment.

Our team's journey has been marked by significant growth and development. We've successfully navigated challenges and fostered a positive team dynamic, as evidenced by our improved score in Healthy, Fact-Driven Conflict. We initially anticipated potential conflicts, and in this case we would solve any major disagreement with a coin toss. Despite our preparations, our team has demonstrated a remarkable ability to resolve issues through open dialogue and mutual respect.

However, we recognize the need to further optimize our approach to Exploitative Learning. By carefully considering individual strengths and providing opportunities for skill development, we aim to maximize our collective potential. Through continued reflection and adaptation, we are confident in our ability to achieve even greater success. The action we will take to resolve the decrease in score for exploitative action will be to more clearly articulate the skills required for a specific task. In addition, it will be necessary to encourage team members to take ownership of their own skills so that we can efficiently determine who should be taking on different tasks.

7 Ethics & Diversity

As engineers, it is important to reflect on the ethical decisions in developing the project, and on how diverse contributors have helped make this device a reality. This section summarizes our thoughts and reflections on the ethical ramifications of the project subsystems and mechanics. For a full treatment, see appendix C.

7.1 Ethics

The conversion device has three primary subsystems:

USB to RGMII Bridge - This system takes a USB signal and converts it into a Reduced Gigabit Media Independent Interface (RGMII) signal. It also receives RGMII data and converts it into USB. This system is primarily responsible for handling sending and receiving data to/from the user's computer.

RGMII to Automotive Ethernet Conversion - This system takes an RGMII signal, converts it into an Automotive Ethernet signal and vice versa. This system is primarily responsible for handling sending and receiving data to/from the car.

Power Subsystem - This system takes the 5V power bus from the usb signal and lowers the voltage to 3.3V to power the data conversion subsystems.

With these subsystems and the device's use-case in mind, we explored different ethical decisions and issues that may arise with the usage of this device, which divide mostly into security concerns and consumer concerns.

Interacting with Safety Critical Functions

Applies to USB to RGMII Bridge and RGMII to Automotive Ethernet Conversion Subsystems

Using the translation device, a bad actor could potentially send commands to safety critical functions (such as steering and brakes) or even rewrite their internal firmware. Additionally, autonomous pathing algorithms and sensor drivers could potentially be tampered with, leading to large scale compromise of the Zoox Vehicle.

User Privacy Concerns

Applies to USB to RGMII Bridge and RGMII to Automotive Ethernet Conversion Subsystems

The adapter also has the potential to be used to snoop the internal communication in the Zoox vehicle. While some of the data may be harmless, a bad actor could potentially scrape location data, rider behavior, or car behavior. With enough data, it could be used for blackmail and impersonation of Zoox's customers.

Sabotage

Applies to the Power Subsystem

If the adapter was tampered with, the power subsystem could potentially be rewired into the communication lines of the car. This could destroy internal electronics or cause undefined behavior long-term.

Planned Obsolescence

Applies to USB to RGMII Bridge and RGMII to Automotive Ethernet Conversion Subsystems

Since the adapter consists of Consumer off the Shelf (COTS) components, long term maintainers of the adapter design are beholden to the chosen component manufacturers. If a manufacturer chooses to end production or software support, maintainers will have to decide whether to redesign the system, or continue using it. Lack of software support may open the user up to security exploits.

Intellectual Property Concerns

Applies to USB to RGMII Bridge and RGMII to Automotive Ethernet Conversion Subsystems

Giving a user access to the entire Automotive Ethernet network in a Zoox car could allow a bad actor to illegally acquire Zoox’s intellectual property. Since this adapter will primarily be internally distributed, this scenario would likely be part of a larger corporate espionage scheme.

7.2 Diversity & Inclusion

It is important to reflect and acknowledge the contributions of past work to current projects, and recognize members of marginalized groups who have contributed greatly to the design and subsystems of this device.

Ajay Bhatt’s pioneering work on USB technology directly impacts our USB 3.0 to Automotive Ethernet adapter project by highlighting the importance of simplicity, universality, and open standards in design. His contributions to creating a reliable, high-speed, and widely compatible interface are foundational to the USB 3.0 standard, which our project depends on for seamless data transfer at up to 5 Gbps.

Bhatt’s work demonstrates how innovative engineering can address universal connectivity challenges, a principle that guides our own design process, regardless of their background. His story also underscores the value of diverse perspectives in engineering, reminding us of the collaborative effort behind global standards. By building on Bhatt’s legacy, our project aims to create a solution that embodies the same spirit of innovation and interoperability that made USB transformative. For a full article, see appendix D.

8 Conclusion

To conclude this document, we will discuss how this project will evolve in the future and how the completion will impact engineering focused industries in the future.

This project contributes to the automotive industry at an engineering and manufacturing level. Companies that develop cars and autonomous vehicles will find a use with this product as it allows technicians and engineers to easily interface with internal networks that use the Automotive Ethernet protocol to send information.

8.1 Future Plans & Testing

As discussed previously, we are currently pursuing two versions of the same device. Version #1 (see section 5.1) is much closer to completion than version #2 (see section 5.2) and therefore we will continue to pursue both in the future, even though version #2 is technically more optimal.

For the first version, the next steps are to complete the PCB layout to transfer the Ethernet signal to Automotive Ethernet. This will then need to be manufactured and tested.

For the second version, the next steps are to complete the schematics and PCB layout for the entire device. The schematics in KiCAD are in progress and will be complete early in the new year. The PCB layout complex and will take some time to complete. Once the board is manufactured it will need to be verified.

Testing and verification for this device breaks into two categories: functionality and performance. To test for device functionality we need to test for correct data transfer at the desired Gigabit speeds. This means we need to analyze incoming and outgoing signals to make sure both carry the correct data and are protocol-compliant. Performance testing entails pushing the device to its limits in order to provide a better description of the device capabilities. We would aim to answer questions such as: when would the device overheat? What is maximum data transfer speed? And many more about the device capabilities.

8.2 Project Resources

This section contains all resources used during development including software tools and hardware.

- **1 GHz Oscilloscope**
 - Essential for observing and analyzing high-frequency automotive Ethernet signals.
 - Enables assessment of signal integrity, timing accuracy, and identification of any interference or noise affecting data transmission.
- **Automotive Ethernet (AE) Signal Test Bench**
 - Allows emulation of real-world automotive communication environments for testing.
 - Assesses the adapter's resilience under varied conditions, such as voltage fluctuations and interference, ensuring compatibility with automotive network protocols.
- **PCB Manufacturing Services**
 - Provides custom printed circuit boards meeting automotive-grade standards for durability and reliability.
 - Ensures proper integration of signal conditioning components and robust routing for AE and USB data lines, maintaining signal strength.

- **Assorted Chips**
 - Includes necessary components like USB3.0 to RGMII chips
 - Potential additional analog and digital components for signal conditioning, power regulation, and noise filtering.
 - See references [10–12, 15, 16]
- **Notion Organization**
 - Provides easy organization for the project documentation
 - Used to organize tasks and calendars for better project management
- **KiCAD 3.0**
 - Software tool used for developing PCB layouts and low level schematics

8.3 Acknowledgements

We would like to acknowledge the contributions of our contact at Zoox, Austin Hendrix, for mentoring us and aiding us through this project. We would also like to acknowledge the help of Dr. John Oliver, and Dr. Simon Tan at Cal Poly for helping us locate testing equipment. We would like to thank the Cal Poly Computer Engineering Department for providing funding for this project.

References

- ¹IEEE, “Ieee standard for ethernet amendment 4: physical layer specifications and management parameters for 1 gb/s operation over a single twisted-pair copper cable”, IEEE 802.3bp (2016).
- ²*Universal serial bus 3.0 specification*, Hewlett-Packard et al. (2008).
- ³A. Ismail, C. Whitby-Strevens, and S. Wallick, “Usb type-c bridge class specification”, (2017).
- ⁴D. Bollati, B. Korber, and M. Kaindl, “Ieee 1000base-t1 system implementation specification”, version 1.6, Open Alliance (2022).
- ⁵*Usb-c to 2.5 gigabit ethernet adapter*, <https://www.cablematters.com/pc-1286-125-usb-c-to-25-gigabit-ethernet-adapter.aspx>, Cable Matters ().
- ⁶*Ae-1000 automotive ethernet adapters*, <https://www.accuratetechnologies.com/Products/Automotive-Ethernet-Adapters>, Accurate Technologies ().
- ⁷*Ieee 1000base-t1 emc test specification for esd suppression devices*, Open Alliance (2020).
- ⁸*Pcb design guidelines for automotive ethernet*, NXP (2023).
- ⁹*Ethernet phy pcb design layout checklist*, Texas Instruments (2021).
- ¹⁰*Lan7800 superspeed usb 3.1 gen 1 to 10/100/1000 ethernet controller*, tech. rep. (Microchip Technology Inc., 2021).
- ¹¹*4-kbit microwire compatible serial eeprom*, tech. rep. (Microchip Technology Inc, 2022).
- ¹²*Dp83867e/is/cs robust, high immunity, small form factor 10/100/1000 ethernet physical layer transceiver*, tech. rep. (Texas Instruments, 2015).
- ¹³*Reduced pin-count interface for gigabit ethernet physical layer devices*, tech. rep. (Hewlett-Packard Company, 2000).
- ¹⁴Y.-C. Chu, *Serial-gmii specification*, tech. rep. (Cisco Systems, 2005).
- ¹⁵*Lan7801 superspeed usb 3.1 gen 1 to 10/100/1000 ethernet controller with rgmii*, tech. rep. (Microchip Technology Inc, 2022).
- ¹⁶*Dp83tg720s-q1 1000base-t1 automotive ethernet phy with sgmmii and rgmmii*, tech. rep. (Texas Instruments, 2020).

A Concept Generation Details

By the time we reached the concept generation phase of our project, we had already developed a preliminary solution for the problem presented by Zoox. In this phase, however, we focused on exploring a broader range of ideas that could potentially address the issue in innovative ways. To facilitate this process, we leveraged ChatGPT’s brainstorming capabilities to generate fresh concepts. Each team member then selected one of these new ideas to research further, evaluating its feasibility for solving the problem. Our research included analyzing the pros and cons of each approach, estimating costs, and documenting additional considerations that might inform our final direction.

Nominal Group Technique:

Each of us used AI to generate a few ideas for the implementation of this device. We then went around the table and discussed the many ideas presented by each person. Each design idea was presented by one person and then discussed by the full group to narrow down the amount of ideas we wanted to consider. (See “Nominal BRAINSTORM” section below).

Yes And:

After going around and narrowing down the ideas we looked at each idea individually to try and elaborate on the methods and processes involved in each approach. The Yes And technique was useful in this case because it was good to bounce off each other for the ideas presented by the AI. This resulted in the four main ideas: FPGA, SoC, Cascade and embedded. (See the “yes and brainstorm” section below). Afterwards we decided that each person would tackle one of the four ideas and try to garner details, pros, and cons about the implementation. This would allow us to conduct productive research for our next meeting.

SCAMPER:

Able to generate our final 4 ideas: FPGA, SoC, Cascade, and embedded. We found that the SoC and embedded solutions could be combined into a singular solution since they shared most of the same implementation details. The FPGA solution would be infeasible to do without more time, so combining the cascaded solution with the FPGA solution resulted in a more feasible concept. Finally, the cascaded adapter solution was originally architected to be a series of three conversion chips, but we discovered that chips exist that can cover two steps of the conversion process at once, simplifying the overall design and adapting it to be superior to the initial pitch.

List of Generated Ideas:

1. Protocol Bridge IC with USB Interface

- Concept: Use a dedicated Ethernet-to-USB bridge IC capable of supporting the specific Automotive Ethernet variant. This IC would handle the conversion of Ethernet frames to USB packets.
- Implementation:
 - Ethernet PHY: Use a compatible PHY, such as an A2B or 1000BASE-T1 PHY, for the Automotive Ethernet input.
 - Microcontroller/FPGA: Integrate an MCU or FPGA with a high-speed USB interface (USB 3.0 or higher) to manage protocol conversion.
 - Power Management: Since Automotive Ethernet often runs at 12V or 24V, add a voltage regulator to convert to 5V and 3.3V for USB and PHY requirements.

- Challenges: Challenges: Protocol and timing management, high data rates, and meeting USB 3.0 compliance.

2. FPGA-Based Converter with Soft IP Cores

- Concept: Leverage an FPGA with Ethernet and USB 3.0 soft IP cores to manage the frame and data conversion.
- Implementation:
 - FPGA Setup: Program the FPGA with custom firmware that interprets Automotive Ethernet frames and re-packets them for USB 3.0.
 - PHY Integration: Use dedicated Automotive Ethernet and USB 3.0 PHYs connected to the FPGA's I/O.
 - Data Mapping: Implement data buffering to handle data rate mismatches and ensure stable throughput.
- Challenges: Complex design and high power consumption, requiring careful thermal and power management.

3. Microcontroller with Dual Protocol Support

- Concept: Utilize a microcontroller (MCU) that supports both Ethernet and USB 3.0, if available, to convert Automotive Ethernet frames directly to USB packets.
- Implementation:
 - MCU Selection: Choose an MCU with built-in Ethernet and USB support, capable of handling the bandwidth of Automotive Ethernet (1 Gbps for 1000BASE-T1).
 - Firmware Development: Develop firmware for frame parsing, conversion, and error-checking to reformat data for USB transmission.
 - Power Management: Integrate with a DC-DC converter to support automotive power levels (12V or 24V).
- Challenges: Limited availability of high-performance MCUs capable of such speeds and complex firmware requirements.

4. Ethernet to USB Adapter with Real Time Operating System

- Concept: Use an RTOS on an embedded processor to manage the high-speed data transfer and protocol conversion efficiently.
- Implementation:
 - RTOS-Driven MCU: Run an RTOS on an MCU capable of high-speed data management (e.g., ARM Cortex-M7 or similar).
 - Ethernet and USB PHYs: Connect separate PHY chips for Automotive Ethernet and USB 3.0, interfaced via GPIO or serial buses.
 - Data Handling: Use RTOS to manage priority tasks, buffer data, and convert packets in real-time.
- Challenges: Timing and synchronization, as USB 3.0 data rates require efficient task handling, and there may be data rate mismatches.

5. Dedicated SoC With Automotive Ethernet Support

- Concept: Utilize a System-on-Chip (SoC) with native support for both Automotive Ethernet and USB, if available.
- Implementation:
 - SoC Selection: Source an SoC designed for automotive applications with support for both 100BASE-T1 or 1000BASE-T1 and USB 3.0.
 - Direct Protocol Handling: Program the SoC to translate and reframe Ethernet packets for USB directly without external components.
 - Enhanced Power Design: Integrate automotive-grade power management for robust operation.
- Challenges: Such SoCs are limited and may not support USB 3.0 natively; high cost and development effort required.

6. Cascading Adapters

- Concept: Use a two-stage conversion: first convert Automotive Ethernet to standard Ethernet, then convert Ethernet to USB 3.0.
- Implementation:
 - Ethernet PHY: Use a PHY to convert Automotive Ethernet (1000BASE-T1) to standard Ethernet (1000BASE-T).
 - USB 3.0 Adapter: Connect a standard Ethernet-to-USB 3.0 adapter to handle the Ethernet packets and reformat for USB.
 - Modular Design: Separate conversion stages, making the setup easier to troubleshoot and potentially more reliable.
- Challenges: Increased complexity and cost due to multiple stages; may introduce latency issues.

Brainstorming Sessions:

- **Protocol Bridge IC with USB Interface:** Utilize a specialized Ethernet-to-USB bridge IC to handle direct protocol conversion between Automotive Ethernet and USB 3.0, simplifying data translation. This approach is cost-effective but limited by the IC's fixed data handling capacity and protocol settings.
 - PROS:
 - * Hardware acceleration ensures that the solution will have minimal latency
 - * Most chips are relatively inexpensive and widely available
 - * Mirrors the client's current adapter setup
 - CONS:
 - * Requires multiple chips to communicate and play nicely
 - * High speed PCB manufacture requires significant knowledge of the board stack and communication protocols
 - * Messing up a design means having to remanufacture and wait for lead times
 - Consideration—Many chips need to read memory that must be externally programmed such as an EEPROM.

- **FPGA-Based Converter with Soft IP Cores:** Leverage an FPGA with custom Ethernet and USB 3.0 IP cores to convert and manage high-speed data in real-time. This flexible solution can handle high data rates but demands complex development and higher power.
 - CONS:
 - * Expensive
 - * difficult to design, need to learn a lot
 - PROS:
 - * Relatively simple conceptually
 - * Easy to test and verify the design
 - * We design the hardware (no reliance on other chips)
 - * Most likely ~\$100
 - Possible Parts:
 - * Artix 7 FPGA (XC7A35T-1FGG484I)
 - * USB Phy/Transceiver (TUSB1310)
 - * Auto Ethernet PHY (DP83TG720S-Q1)
 - * EEPROM style memory needed
- **Ethernet-to-USB Adapter with Real-Time Operating System (RTOS):** Run an RTOS on an MCU to prioritize and manage high-speed data tasks for seamless Ethernet-to-USB conversion. This setup is efficient and adaptable but requires intricate firmware to maintain USB 3.0 speeds.
 - PROS: enables precise timing and scheduling, which is valuable for applications requiring low latency and real-time data transfer.
 - CONS:
 - * Availability of MCUs that support both protocols with an RTOS at USB 3.0 speeds is limited.
 - * High-performance RTOS licenses can be costly, which may not be suitable for low-budget or low-margin projects.
 - * Some RTOSs may increase power usage due to the need for continuous task scheduling and resource management.
- **Dedicated SoC with Automotive Ethernet and USB Support:** Use an SoC that natively supports both protocols, simplifying integration and data handling. However, compatible SoCs are costly and limited, with few customization options available.
 - PROS:
 - * Mostly software so has the potential to be reusable.
 - * Adapter could do more than just translate (maybe even interpret?)
 - CONS:
 - * SoCs with both Automotive Ethernet and USB 3.0 support are rare and typically expensive.
 - * SoCs are incredibly hard to implement on a board (see cubesat systemboard)

- * Unless we are using onboard hardware on the SoC, achieving max speed is completely reliant on the CPU clock speeds.
- * Power hungry – using a general purpose processor will require external power
- Since this solution breaks two of our major requirements, it is not feasible to implement.
- **Automotive Ethernet to Standard Ethernet, Then Ethernet-to-USB Adapter:** Convert Automotive Ethernet to standard Ethernet first, then connect to an Ethernet-to-USB adapter for USB 3.0 output. This modular design is flexible but introduces additional latency and complexity. CONS: Dual conversion stages add latency and may not be suitable for applications needing real-time data transfer.
- **Custom Embedded Solution:** For custom needs or prototyping, you could design an embedded system, such as an ESP32 or Raspberry Pi with an automotive Ethernet PHY, to read the automotive Ethernet and output data through USB.
 - PROS:
 - * Ability to integrate additional sensors or peripherals, such as GPIO, ADC, or Wi-Fi/Bluetooth modules for enhanced functionality.
 - * Using off-the-shelf hardware like the ESP32 or Raspberry Pi can be more cost-effective for initial development and prototyping.
 - CONS:
 - * This approach would require writing custom firmware to translate the Ethernet frames and might require significant development time.
 - * Increased complexity in system design and implementation, requiring expertise in embedded systems, networking, and firmware development.
 - * Depending on the platform chosen, size and power consumption may not meet the requirements for certain automotive applications.
 - * Scaling the solution for mass production may require additional engineering efforts and could lead to higher costs compared to off-the-shelf solutions.
 - IMPLEMENT:
 - * Raspberry Pi 4: \$35
 - * PHY: \$5-6
 - Gigabit ethernet capabilities
 - USB3.0
 - RAM options: 1,2,4,8 GB
 - Adding a PHY like the **DP83TC811** can allow the Raspberry Pi to interface with 100BASE-T1 or 1000BASE-T1 automotive Ethernet
 - DP83TC811R-Q1: Low-power 100BASE-T1 automotive Ethernet physical layer transceiver PHY (RGMII Interface)
 - DP83TG720R-Q1: 1000BASE-T1 automotive Ethernet PHY (RGMII interface)
 - * CONSIDERATIONS:
 - has USB 3.0 ports, which are theoretically capable of supporting gigabit speeds (up to 5 Gbps). However, the real-world throughput is typically somewhat lower due to processing overhead.

- The built-in Ethernet port on the Pi 4 supports 1 Gbps speeds, but it is connected internally through a shared PCIe bus along with USB, which can introduce bandwidth limitations under heavy load.
- The Raspberry Pi 4 does not have a native RGMII or GMII interface, which complicates direct connectivity. You may need an RGMII-to-SGMII converter to bridge the PHY to the Pi, or consider an intermediary processor or FPGA to handle the conversion at gigabit speeds.
- Even if a physical connection is established, the Raspberry Pi 4's ARM CPU might struggle to handle full-duplex gigabit speeds, especially if protocol translation or heavy data processing is involved.
- the software stack would need efficient drivers and possibly custom code to minimize latency.

B ITP Metrics

There were two ITP Metrics reports during this project so far. Each team report features four main categories in which the team is evaluated. These categories are

1. Communicate: how well the team works and speaks to each other, and how efficiently
2. Adapt: How well the team changes based on the current situations and how we efficiently reach our goals
3. Relate: How well the team works together and helps each other
4. Educate: How much the team members learn from one another and how much we utilize all the skill sets present in the team

B.1 ITP Metrics #1 Team Report

Wednesday, October 23rd, 2024

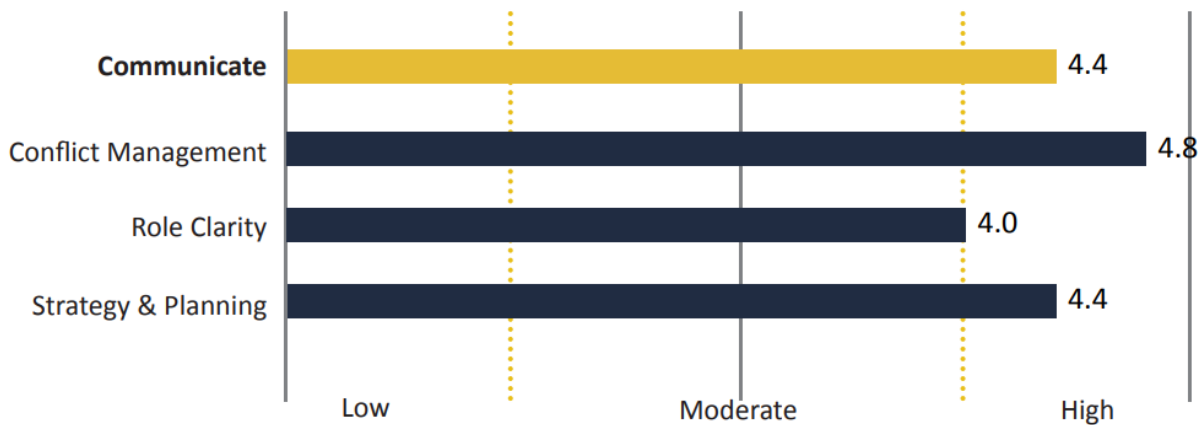


Figure 5: First ITP Metrics Team Report - Communication

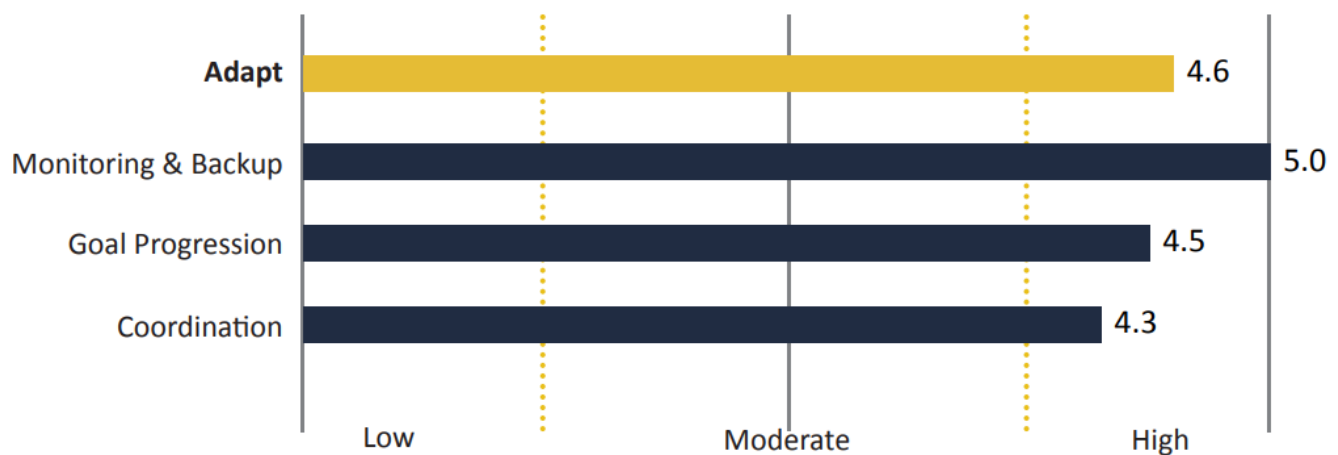


Figure 6: First ITP Metrics Team Report - Adapt

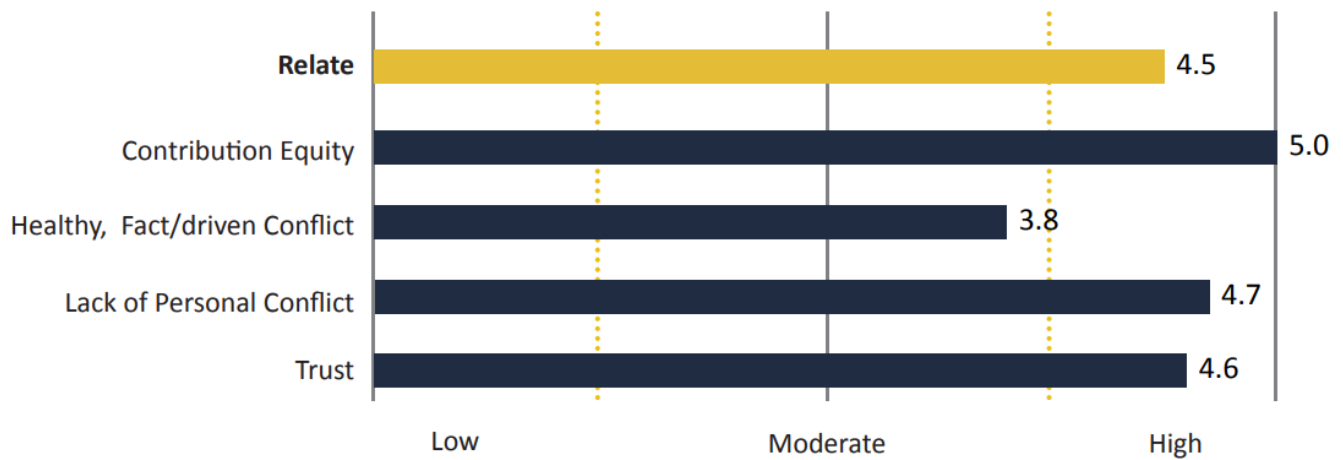


Figure 7: First ITP Metrics Team Report - Relate

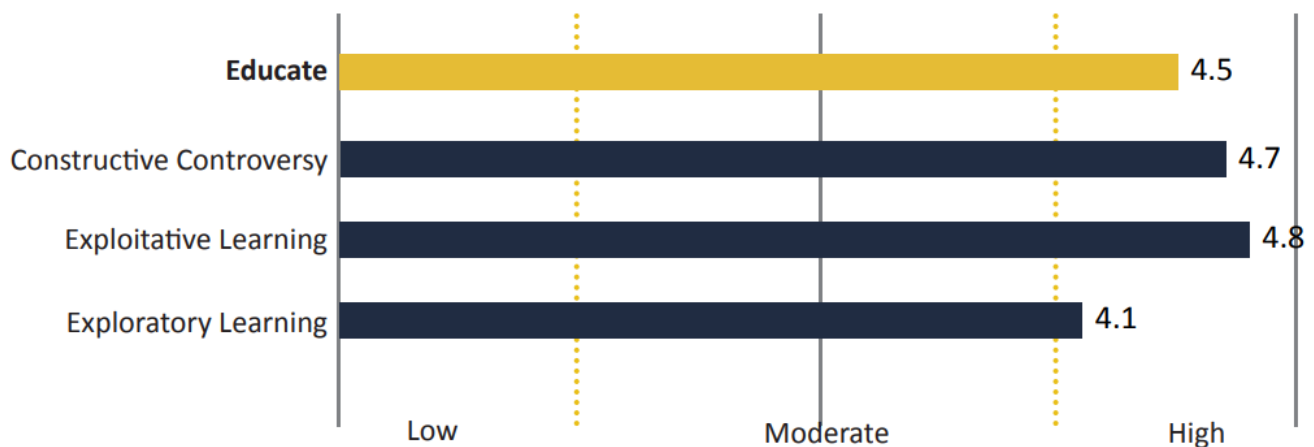


Figure 8: First ITP Metrics Team Report - Educate

Peer Review Reflections (Individual):

Alex - My peer review ratings were mostly positive. The one place that could use the most improvement is the 'focus' category. I will try and make sure that this project is a priority and I will always try and dedicate a good amount of time to this project.

Vlad - For the most part, my self rating was fairly close to my peer rating, but the communication and focus categories can be improved.

Seth - In reflecting on the recent self-assessment where I consistently gave myself lower scores in each section than what my team members gave me, I realize that I need to stop undervaluing my own skills and contributions.

Ben - Overall, my peer's assessment correlated well with my personal scores. Although still strong, I could improve my focus during group meetings.

Team Dynamic Reflections (Group):

Healthy, Fact/driven Conflict

Our score for this section is a 3.8. This is not technically a blind spot, however, we came to the conclusion that strategizing for this scenario is important. In the case that there is a disagreement between two team members, we will have the other two unbiased members discuss the options and make the final decision. If the issue involves everyone, we can attempt to make use of design assessment tools like charts and tables that help assess the ‘quality’ of each idea from a potentially more objective point of view, and if all else fails, we will flip a coin.

Exploratory Learning

To gain a deeper understand of of all aspects of the project, we have designated reading protocols such as USB3.0, AutomotiveEthernet, RGMII/SGMII, and Ethernet to each of our group members, who in turn compiled necessary the information from the protocol documents to share with our group. Subsequently, we are utilizing the concept generation exercise to kickstart out of the box thinking alongside deeper understanding of potential solutions that can tackle our problem. These two exercises should not only increase our exploratory learning.

B.2 ITP Metrics #2 Team Report

December 4th, 2024

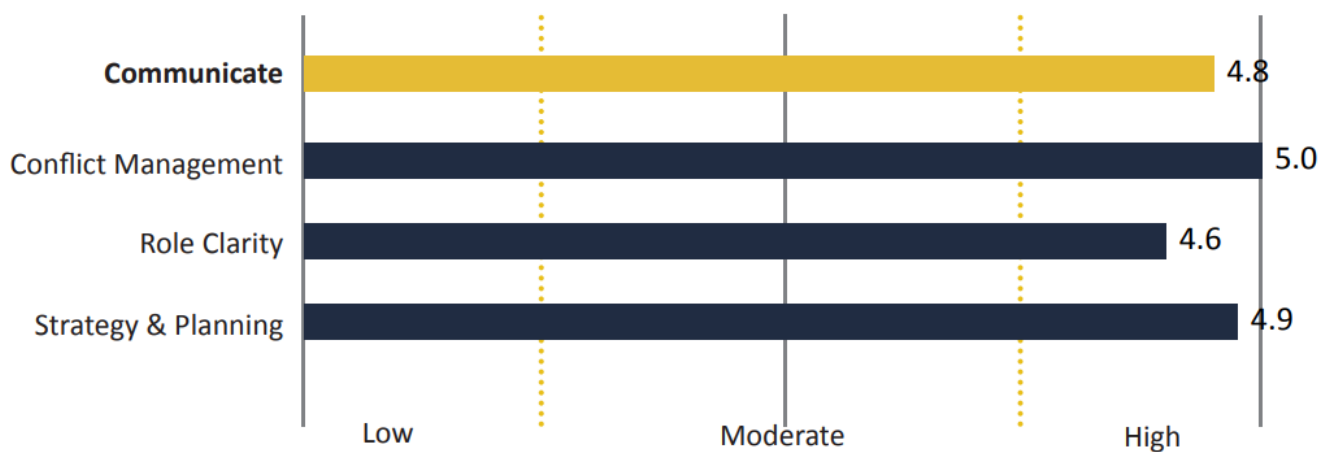


Figure 9: Second ITP Metrics Team Report - Communication

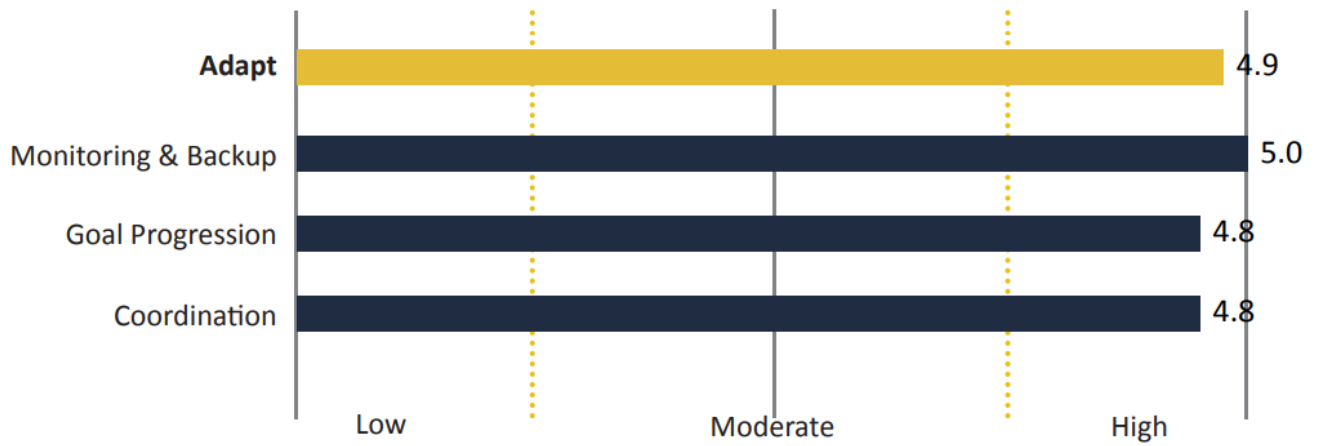


Figure 10: Second ITP Metrics Team Report - Adapt

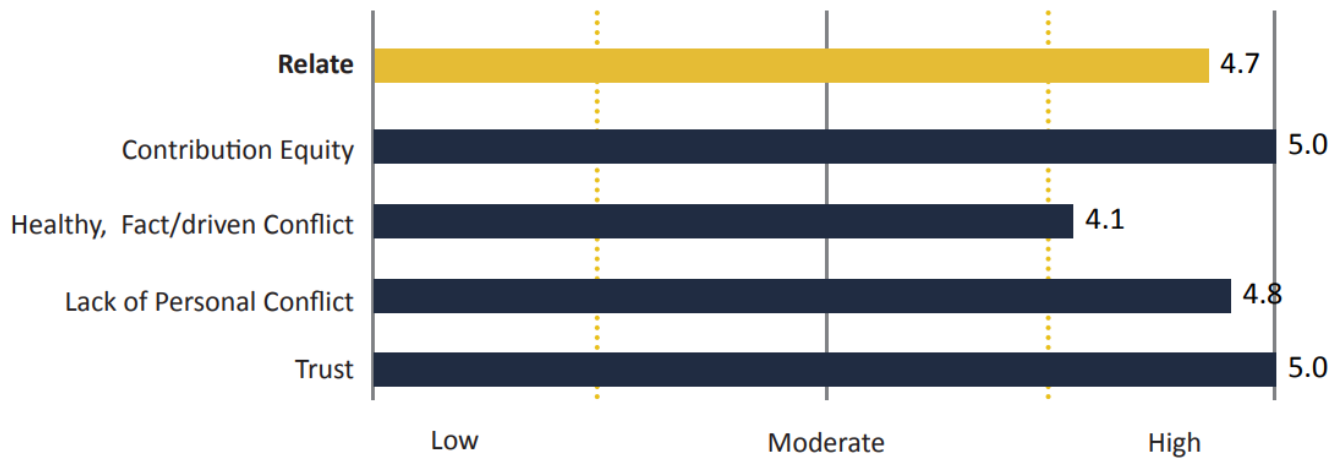


Figure 11: Second ITP Metrics Team Report - Relate

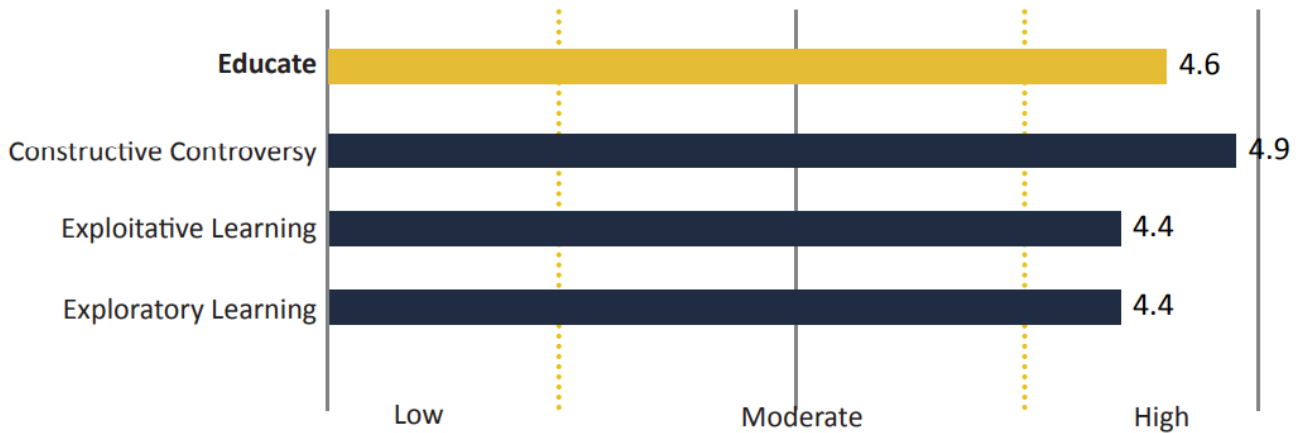


Figure 12: Second ITP Metrics Team Report - Educate

Peer Review Reflections:

Alex - My peer rating has improved greatly, with seemingly no room for improvement in all the categories. Personally, I think I can still improve in the communication and focus categories, and I will try and do so in the future.

Vlad - My peer rating has improved or stayed the same in all categories. My peer rating is within 0.3 points of my self rating for all categories except communication. That being said, the communication peer rating has improved since last time.

Seth - After addressing my tendency to undervalue my contributions, my latest self-assessment was more aligned with my team's feedback. This change has boosted my motivation, reinforcing the importance of adjusting your perspective.

Ben - Looking at my next personal assessment, my self-rating correlated well with my peers's rating. Overall, all of my scores improved versus the last evaluation round. I think as the project has taken shape, my focus on the task at hand has improved. The two items that I can still improve on slightly are my capabilities and commitment. I think by being more involved in the design in CPE 450, both of these scores will improve.

Team Dynamic Reflections:**Healthy, Fact-Driven Conflict**

Our score in this category improved from 3.8 to 4.1, showing progress. While we discussed conflict resolution strategies in our last review, we haven't had to use them yet. This suggests that our concerns about group conflict may have been unnecessary, as we've been able to resolve issues through fair and open discussions without resorting to strategies like coin flips. This is a positive sign of our team's ability to work together effectively.

Exploitative Learning

Our score in this category decreased slightly from 4.8 to 4.4. While still strong, this drop shows we need to focus on using each team member's strengths more effectively. To improve, we plan to slow down task delegation and spend more time discussing who is best suited for each task. At the same time, we want to ensure that everyone has opportunities to learn new skills. Balancing efficiency with skill development will help us get back on track in this area.

C Detailed Ethics Reports

C.1 Vladimir

There are two main hardware subsystems for this project:

1. The first subsystem converts USB into standard ethernet.
2. The second subsystem converts standard ethernet into automotive ethernet.

Since MicroChip provides drivers for the LAN7800 chip, and also provides tools for creating EEPROM images and writing them to the EEPROM, we do not necessarily have to develop our own software for this project. Therefore, I do not consider our project to have any software subsystems.

For the USB to standard ethernet portion, possible issues include worsening a digital divide, material sourcing, planned obsolescence, driver vulnerabilities, and trade regulation. For the standard ethernet to automotive ethernet portion, possible issues include lack of familiarity, material sourcing, planned obsolescence, hardware/firmware vulnerabilities, and trade regulation. The two hardware subsystems evidently share some possible issues.

I consider the importance of these issues to be in the following order:

1. Driver vulnerabilities
2. Hardware/firmware vulnerabilities
3. Planned obsolescence (for the USB side)
4. Planned obsolescence (for the ethernet side)
5. Material sourcing (for the ethernet side)
6. Material sourcing (for the USB side)
7. Trade regulation (for the ethernet side)
8. Trade regulation (for the USB side)
9. Lack of familiarity
10. Worsening a digital divide

The reason why I chose this order is because realistically, practically every piece of software has at least some sort of bug, and since device drivers are often more privileged than userland software, a vulnerability in a driver can be very bad. The next possible issue is hardware/firmware vulnerabilities. These are problematic because once a device has been sold, it's very unlikely that an end user could fix something this low-level. The next possible issue is planned obsolescence. A manufacturer could stop releasing drivers for newer versions of operating systems in order to "encourage" people to buy new devices instead. Extracting materials that are needed for producing these adapters could be bad for the environment. There could also be trade restrictions between certain countries. There could be a lack of familiarity with automotive ethernet for many people. These sorts of adapters could worsen a digital divide.

In my opinion, the top three possible ethical or security issues that a USB-ethernet adapter could have are:

1. Driver Vulnerabilities
2. Hardware/firmware vulnerabilities
3. Planned obsolescence.

The reason why I chose this order is because realistically, practically every piece of software has at least some sort of bug, and since device drivers are often more privileged than userland software, a vulnerability in a driver can be very bad. The next possible issue is hardware/firmware vulnerabilities. These are problematic because once a device has been sold, it's very unlikely that an end user could fix something this low-level. The next possible issue is planned obsolescence. A manufacturer could stop releasing drivers for newer versions of operating systems in order to "encourage" people to buy new devices instead. Although a USB to ethernet adapter doesn't seem like it could be questionable in any way, it could potentially have some problems.

C.2 Seth

There are a select few issues with our project but despite the minimal quantity of issues, some problems are going to be very difficult to solve. The main problem that we've encountered is the extreme difficulty in testing our product. This might be categorized as a societal issue because the equipment required to test or adapter is extremely hard to come by as a result of its exorbitant price tag. This will result in extreme limitations on access to individuals without a large amount of capital to use on testing devices. For this project, we need an oscilloscope/logic analyzer with the ability to measure gigabit frequency (as used by Automotive Ethernet), which has a market value of \$20,000-\$30,000.

Another high-level issue we might face with our project stems from its potential contribution to the autonomous vehicle industry. While autonomous vehicles promise advancements in safety and efficiency, they also raise significant societal and ethical concerns. Chief among these is the potential reduction of jobs in industries heavily reliant on human drivers, such as trucking, delivery, and ride-sharing services. By contributing to the development of such technologies, our work inadvertently amplifies this risk, necessitating careful consideration of the broader social implications of our progress. Balancing technological advancement with the preservation of livelihoods is a challenge that demands attention throughout the project's development.

Finally, the adapter introduces several potential security challenges that must be carefully addressed to ensure the safety and reliability of the system. One significant concern is the possibility of unauthorized access or tampering with the device. Since the adapter interfaces with critical automotive systems, vulnerabilities in its firmware or drivers could allow attackers to intercept or manipulate data transmitted via Automotive Ethernet. Such an exploit could compromise vehicle safety or expose sensitive information, making robust encryption and authentication mechanisms essential.

Another security issue arises from the use of EEPROM for storing configuration data. If proper protections are not in place, malicious actors could alter stored parameters, potentially leading to device misbehavior or disruption in communication. Additionally, the adapter's compatibility with various operating systems, including Windows, Linux, and macOS, introduces the risk of platform-specific vulnerabilities, which could be exploited to inject malware or disable the device remotely. Addressing these risks requires rigorous security testing, regular firmware updates, and adherence to secure design principles throughout development.

SELECTED SUBSYSTEMS:

USB Interface: Handles USB 3.0 connections to host devices.

RGMIIBridge: Converts USB to RGMII signals.

Power Management: Ensures proper voltage levels and current stability for the ICs.

EEPROM: Storage for the device's configuration

C.2.1 USB Interface

Potential Problems or Issues

1. **Compatibility:** The interface may not work seamlessly across all operating systems (Windows, Linux, macOS) or with certain host devices.
2. **Signal Integrity:** High-speed data transfer might suffer from signal degradation, skew, or jitter, especially over longer cables or suboptimal PCB traces.
3. **Power Delivery:** USB-C power requirements may vary, and inadequate design could lead to power failures or overheating.
4. **Security:** USB interfaces are vulnerable to attack vectors like unauthorized data access or malicious firmware injection.
5. **Certification:** Failure to comply with USB-IF standards can result in operational issues and market restrictions.

C.2.2 RGMII Bridge

Potential Problems or Issues

1. **Timing Mismatches:** Inadequate timing alignment between clock and data signals can cause communication errors.
2. **Electromagnetic Interference (EMI):** High-speed RGMII signals may cause or be affected by EMI, leading to degraded performance.
3. **Signal Loss:** Poor PCB design or trace layout could result in significant signal loss or noise.
4. **Latency:** Suboptimal protocol translation might introduce delays, affecting real-time communication.
5. **Interoperability:** Compatibility issues with downstream Automotive Ethernet PHYs or upstream USB interfaces could hinder overall performance.

C.2.3 Power Management

Potential Problems or Issues

1. **Voltage Instability:** Voltage fluctuations can damage sensitive ICs or cause erratic behavior in the system.
2. **Overheating:** Inefficient power management or poor thermal design might lead to excessive heat generation.
3. **Power Sequencing Failures:** Incorrect sequencing could cause ICs to power up in an unstable state, leading to potential damage.

4. **Efficiency Losses:** Poor power regulation could result in unnecessary power dissipation and reduce overall system efficiency.
5. **Noise and Ripple:** Electrical noise in power lines can interfere with high-speed data processing, particularly in the USB and RGMII subsystems.

C.2.4 EEPROM

Potential Problems or Issues

1. **Data Corruption:** Power loss during write cycles can lead to corrupted configuration data.
2. **Write/Erase Cycles:** Exceeding the EEPROM's maximum endurance can cause permanent data storage failures.
3. **Unauthorized Access:** Lack of proper security measures could allow malicious reconfiguration of critical settings.
4. **Integration Errors:** Misconfigured or incompatible settings could prevent proper device initialization.
5. **Limited Storage:** Insufficient capacity to store future updates or additional configurations may require hardware redesigns.

REFLECTION:

This project provided me with a deeper understanding of the intricate balance between technical design and broader societal considerations. Identifying and analyzing societal, environmental, ethical, security, and political issues revealed how interconnected engineering projects are with both global and local contexts. It allowed me to see how decisions at the subsystem level, such as ensuring robust security measures or addressing barriers to accessibility, could have far-reaching implications beyond the technical sphere. This process strengthened my ability to evaluate problems critically and conduct meaningful research. It also emphasized the importance of a holistic approach to engineering, one that prioritizes technical excellence while considering its broader societal impact.

C.3 Benjamin

Subsystems:

1. USB Controller Side - This system takes a USB signal from a computer and converts it into RGMII (reduced gigabit media independent interface)
2. Automotive Ethernet PHY Side - This side takes an RGMII signal and converts it to an automotive ethernet signal

C.3.1 Subsystem 1

When using a USB controller that converts USB to RGMII (Reduced Gigabit Media Independent Interface), the following ethical concerns might arise, depending on the context and application:

1. Data Privacy and Security

- **Interception Risks:** The USB-RGMII converter may introduce vulnerabilities where data transmitted over the network can be intercepted if the device does not employ secure encryption.
- **Unauthorized Access:** If the USB controller allows access to a network, it may be exploited by malicious actors to gain unauthorized entry, leading to data breaches.

2. Device Trustworthiness

- **Malware and Backdoors:** If the USB controller or firmware is compromised or comes from an untrusted source, it might include malware or backdoors that could compromise connected systems or networks.
- **Firmware Manipulation:** Unsigned or modifiable firmware may be exploited to alter device behavior, potentially leading to malicious activity.

3. Regulatory Compliance

- **Network Standards:** Devices that convert USB to RGMII should comply with IEEE standards for Ethernet and network communication. Non-compliant devices might create interoperability or legal issues in regulated environments.
- **Export Control:** Such devices may be subject to export restrictions, especially if they can be used in critical infrastructure or military applications.

4. Intellectual Property

- **Reverse Engineering:** Converters could be used to reverse-engineer proprietary network protocols or hardware, raising concerns about intellectual property infringement.
- **Unlicensed Use:** The use of patented or proprietary technology in the USB controller without proper licensing might breach intellectual property laws.

5. Ethical Deployment

- **Surveillance:** If used in surveillance systems or network monitoring, the device might contribute to unethical uses like unauthorized spying or monitoring.
- **Harmful Applications:** These devices could be used in applications that facilitate censorship, discrimination, or other ethically questionable actions.

6. Environmental Impact

- **E-waste:** If such devices are used and discarded frequently, it contributes to e-waste. Ensuring they are recyclable and disposed of responsibly is crucial.

Mitigation Measures:

- Use controllers from reputable manufacturers with secure firmware and software support.
- Ensure proper encryption protocols are in place for data transmitted via the device.
- Comply with industry standards and regulatory requirements.
- Regularly update firmware to address vulnerabilities.
- Limit usage to ethical applications and monitor for misuse.

By being mindful of these concerns, you can better ensure that such technology is used responsibly and ethically.

C.3.2 Subsystem 2

1. Safety and Reliability

- Automotive Ethernet PHYs are often used in safety-critical systems like Advanced Driver Assistance Systems (ADAS) or autonomous vehicles. Any failure, malfunction, or design flaw could endanger lives. If the PHY is not rigorously tested and validated for the intended use case, it might compromise the safety of the vehicle's operation.

2. Security Risks

- Automotive Ethernet networks connect various systems (e.g., infotainment, braking, powertrain). An unsecured PHY could be exploited by hackers to gain unauthorized access, potentially taking control of vital vehicle functions. Malicious actors could modify the firmware or hardware of the PHY to introduce vulnerabilities or unauthorized behavior.

3. Data Privacy

- Automotive Ethernet networks often transmit sensitive data about the vehicle, driver, and passengers. If the PHY lacks robust security mechanisms (e.g., encryption), it could expose this data to interception or misuse. Data passing through the PHY could also be used for unauthorized monitoring or surveillance of drivers, raising ethical questions about consent and data ownership.

4. Compliance with Standards

- The PHY must comply with functional safety standards like ISO 26262 for automotive applications. Using non-compliant hardware in critical systems is ethically questionable and may violate regulations. Devices should adhere to established Ethernet standards (e.g., IEEE 802.3) to ensure reliable communication. Non-compliance could lead to network instability or accidents.

5. Intended Use and Misuse

- PHYs used in aftermarket systems or non-OEM applications may introduce risks if they are not designed to the same standards as OEM parts. An RGMII-compatible PHY could be used in devices designed to bypass safety or emissions controls, raising ethical and legal issues.

6. Intellectual Property

- Using an automotive Ethernet PHY could facilitate reverse engineering of proprietary automotive network protocols, potentially violating intellectual property rights. If the PHY or its associated firmware contains unlicensed patented technologies, it might infringe on intellectual property laws.

7. Environmental Concerns

- Automotive-grade PHYs are specialized devices. If improperly disposed of, they contribute to electronic waste and environmental harm. If the PHY is not designed for low power consumption, it may reduce vehicle efficiency, contributing to higher emissions indirectly.

8. Ethical Deployment in Critical Systems

- Using the PHY in applications that are not disclosed to end users, such as data collection for insurance purposes or location tracking, can lead to ethical concerns. The PHY could also be repurposed for non-automotive or unethical applications, such as surveillance or creating spoofed vehicle networks for malicious purposes.

C.3.3 Overall Rankings

1. Security and Trustworthiness
2. Privacy
3. Intellectual Property
4. Standards Compliance
5. Environmental Concerns

As a device primarily intended for diagnostic and repair, security is of the top ethical concern for this device. Since the automotive ethernet network interacts with the entire car, a bad actor could theoretically use the device to tamper with sensors, pathing algorithms, speed checking, and potentially safety-critical functions such as steering, acceleration and braking. Autonomous vehicles already stand as a safety question for the general public, making tolerance for malfunction low. Privacy is also an issue since riders could have location, facial data, and usage habits exposed from the car. Intellectual property is of slightly less concern since the intent is for company employees to utilize this adapter. However, if open-sourced, extra care needs to be taken not to unnecessarily expose IP. Since this is an electrical device used to see critical information, it must comply with all current standards. Finally, although still a concern, the environmental impact this device will have should be relatively low due to its low volume of production.

Web Search Reflection:

Already, hackers are finding exploits to gain access into a cars internal systems. Additionally these exploits are often used to spy on unsuspecting drivers. Finally, most of the concern with data exposure relates to car owners, but a law firm website highlights the additional concerns that IP holders may have.

Personal Reflection:

Prior to doing any ethical research, I saw the system we were creating as a passive translator. Now seeing how large ethical concerns relate to our system, I realize that care must be taken to ensure we minimize any negative impact. Since our device exposes an internal communication network, it can expose user data, safety critical systems, and potentially our client's intellectual property.

C.4 Alex

Subsystem Identification:

1. Automotive Ethernet connection and translation
2. USB connection and translation
3. Power control

4. Peripherals and user interface

I will justify this rank ordering by defining each of the subsystems. The first is the connection to the automotive ethernet. This is arguably the most important as it is what delivers the information to whichever vehicle the adapter is connected to. Next, the USB connection, which takes information from the user, is still important, but much easier to implement as USB is very standard in modern society. Third, the power control is the subsystem containing all regulators and inductors which control the various required voltages and power inputs. This subsystem is important because it is necessary at the lowest level of design in order to make a working device. Finally, the peripherals and user interface. I chose this as the lowest rank as it is an addition that will make the user experience better, not one that contributes to the functionality of the adapter.

From web searches the main issues with the subsystems have to do with security and intended use. This adapter is an electronic device, and therefore can interfere with electronics inside the receiving vehicle. If the power module is sabotaged or similar, it can damage much more expensive and important electronics. In addition, the device depends on third party components, meaning components bought from other companies, which could pose a security risk if those companies were to include a hardware trojan or something similar, then potentially secure data could be at risk. Finally, the intended use is important when considering the ethics of this device. It could theoretically be used to develop any vehicle, and vehicles have many use cases, one of which is military, which shows some ethical ramifications, especially as autonomous warfare is becoming more and more probable.

D Diversity, Equity, & Inclusivity

USB Development and Ajay Bhatt’s Role in Shaping Modern Connectivity

For our project, which bridges **Automotive Ethernet**[1] and **USB 3.0**[2], the USB standard is a crucial part of the system. The Universal Serial Bus has become the default interface for connecting peripherals to computers and transferring data. While USB is now very common, its origins are rooted in innovation by a team at Intel, including Ajay Bhatt, whose contributions have often been overshadowed.

Ajay Bhatt, an Indian-born computer engineer, played a pivotal role in the development of USB technology while working at Intel in the 1990s. The USB’s goal was to simplify connectivity and replace a chaotic array of proprietary ports and connectors. This would enable seamless plug-and-play functionality. Bhatt’s team envisioned a universal connector that could simplify how devices communicate and share data across systems.

Despite his groundbreaking work, Ajay Bhatt is often un-recognized in the broader narrative of USB’s success. Marketing campaigns and public attention often credit corporations like Intel or high-profile executives, leaving engineers like Bhatt in relative obscurity. For example, Intel’s branding benefited from USB’s ubiquity, but Bhatt’s role rarely received the attention it deserved outside of technical communities.

Ajay Bhatt’s contributions to USB are directly relevant to our adapter project, which relies on USB 3.0’s high-speed data transfer capabilities. USB 3.0, introduced in 2008, built upon Bhatt’s foundational work on earlier USB versions by dramatically increasing throughput (up to 5 Gbps) and enhancing power efficiency. The reliability, ease of use, and universal compatibility of USB interfaces are essential for our design, ensuring that users can connect a plethora of modern devices to automotive Ethernet networks seamlessly.

Bhatt’s work also exemplifies the importance of engineers from underrepresented backgrounds in shaping global standards. His journey as an immigrant highlights the value of diverse perspectives in engineering, which often lead to solutions that address universal challenges, such as the lack of standardized connectivity in the 1990s.

Ajay Bhatt’s contributions emphasize the value of design simplicity and universality, principles that directly inform our adapter’s architecture. By learning from USB’s development, we recognize the importance of adhering to open standards, ensuring that our adapter supports broad interoperability across devices and platforms.

In honoring figures like Bhatt, our project also aligns with the broader goal of celebrating diverse contributions in engineering. Recognizing his pivotal role underscores the collaborative nature of technological innovation, where many unsung heroes lay the groundwork for transformative progress.